

Carry Out Data Mining and Machine Learning with WEKA



WEKA is a collection of visualisation tools and algorithms for data analysis and predictive modelling. It is very popular among users and developers as can be seen by the large number of downloads.

In the current era of Big Data and enormous data processing tasks, the analysis of huge data sets has become a complex and challenging issue. There are assorted software products in the global market for data interpretation and behaviour analysis, yet the development and enhancements in data mining and machine learning tools is continuously being researched. Data mining, or to put it simply, the knowledge discovery process, is one of the thrust areas for scientists, practitioners as well as research scholars searching for patterns in large data sets involving methods at the intersection of artificial intelligence, machine learning, statistics, and database systems. Data mining extracts information from a large data set and transforms it into a specific structure that can be used for analysis and future predictions. Data mining tasks involve data pre-processing, model and inference considerations, interestingness metrics, complexity considerations, post-processing of discovered structures, visualisation, and online updating.

A number of free and open source software tools are available for data mining and machine learning tasks that can handle data sets of any discipline, whether they are for agriculture, defence, forensics, gaming, business, science or medical data.

WEKA (Waikato Environment for Knowledge Analysis) is an excellent open source software that is widely used and very famous in the research market. Thousands of practitioners and research scholars execute

data mining algorithms on the WEKA tool. WEKA is a globally accepted tool of machine learning software that is developed in Java by the University of Waikato, New Zealand under the GNU General Public License. A compilation as well as collection of machine learning algorithms for data mining tasks has been embedded in WEKA. These data mining and machine learning algorithms can be applied to the dataset of any domain. Besides, the algorithms can be called from its own Java code. The software has a collection of tools for various data mining primitive tasks including data pre-processing, classification, regression, clustering, association rules and visualisation. The software is also well-suited to develop new algorithms for data mining and machine learning.

The official website to download WEKA from is <http://www.cs.waikato.ac.nz/ml/weka>. The tool can be downloaded for multiple operating systems from <http://www.cs.waikato.ac.nz/ml/weka/downloading.html>. Rich documentation is available for users as well as developers. The current stable release of WEKA is 3.6.10. It can also be downloaded from the Web-based source code repository SourceForge.net. The popularity of WEKA can be estimated by the download statistics: 16,000 downloads per week from SourceForge.net [noted from SourceForge.net Download Statistics].

The WEKA user interface

Explorer is the main user interface of WEKA, but the same functionality can be accessed through the